

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

Find $\int 6x^5 dx$.

☐ **A** $6x^6 + C$

☐ **B** $x^6 + C$

☐ **C** $30x^4 + C$

☐ **D** $x^5 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(a) · printed p. 496 / PDF p. 513

Q2 · Rank 1 / Foundation

Find $\int x^4 dx$.

☐ **A** $4x^3 + C$

☐ **B** $x^5 + C$

☐ **C** $x^5/5 + C$

☐ **D** $x^4/4 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(b) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q3 · Rank 1 / Foundation

Find $\int (2x^3 - 6x) dx$.

☐ A $2x^4 - 6x^2 + C$

☐ B $6x^2 - 6 + C$

☐ C $x^4/2 - 6x^2 + C$

☐ D $x^4/2 - 3x^2 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(f) · printed p. 496 / PDF p. 513

Q4 · Rank 2 / Core

Find $\int (x^2 - 8x + 3) dx$.

☐ A $x^3/3 - 4x^2 + 3x + C$

☐ B $x^3 - 8x^2 + 3x + C$

☐ C $x^3/3 - 4x^2 + 3 + C$

☐ D $2x - 8 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(g) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q5 · Rank 2 / Core

Find $\int (ax + b) dx$, where a and b are constants.

☐ **A** $ax^2 + bx + C$

☐ **B** $ax^2/2 + bx + C$

☐ **C** $ax^2/2 + b + C$

☐ **D** $a + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 1(h) · printed p. 496 / PDF p. 513

Q6 · Rank 2 / Core

Find total cost if marginal cost is $MC = Q + 5$ and fixed costs are 20.

☐ **A** $Q + 25$

☐ **B** $Q^2/2 + 5Q$

☐ **C** $Q^2/2 + 5Q + 20$

☐ **D** $Q^2 + 5Q + 20$

Jacques, 9th ed. · §6.1 · Exercise 6.1 2(a) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q7 · Rank 3 / Application

Marginal cost is $MC = 2Q + 6$. Total cost is 212 when $Q = 8$. Find total cost when $Q = 14$.

☐ A 280☐ B 416☐ C 28☐ D 380

Jacques, 9th ed. · §6.1 · Exercise 6.1 3 · printed p. 496 / PDF p. 513

Q8 · Rank 3 / Application

Find the total revenue and demand functions corresponding to $MR = 20 - 2Q$. Use $TR(0) = 0$ and $Q > 0$ for the demand function. Choose (TR, P) .

☐ A $(20Q - Q^2, 20 - Q)$ ☐ B $(20Q - Q^2, 20 - 2Q)$ ☐ C $(20 - 2Q, 20/Q - 2)$ ☐ D $(20Q - 2Q^2, 20 - 2Q)$

Jacques, 9th ed. · §6.1 · Exercise 6.1 4(a) · printed p. 496 / PDF p. 513

Lesson 13 · Antiderivatives and recovering totals

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q9 · Rank 3 / Application

Find the consumption function if the marginal propensity to consume is 0.6 and consumption is 10 when income is 5. Deduce the savings function. Choose $(C(Y), S(Y))$.

☐ A $(0.6Y + 10, 0.4Y - 10)$

☐ B $(0.6Y + 7, 0.4Y - 7)$

☐ C $(0.6Y + 7, 0.4Y + 7)$

☐ D $(0.6Y, 0.4Y)$

Jacques, 9th ed. · §6.1 · Exercise 6.1 5 · printed p. 496 / PDF p. 513

Q10 · Rank 4 / Challenge · OPTIONAL

Differentiate $F(x) = (2x + 1)^5$. Hence find $\int (2x + 1)^4 dx$.

☐ A $(2x + 1)^5/5 + C$

☐ B $2(2x + 1)^5/5 + C$

☐ C $(2x + 1)^5/10 + C$

☐ D $8(2x + 1)^3 + C$

Jacques, 9th ed. · §6.1 · Exercise 6.1 8(1) · printed p. 496 / PDF p. 513